```
!pip install pandas matplotlib seaborn numpy
```

```
Requirement already satisfied: pandas in c:\anaconda3\lib\site-
packages (2.3.3)
Requirement already satisfied: matplotlib in c:\anaconda3\lib\site-
packages (3.10.6)
Requirement already satisfied: seaborn in c:\anaconda3\lib\site-
packages (0.13.2)
Requirement already satisfied: numpy in c:\anaconda3\lib\site-packages
(2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\anaconda3\
lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\anaconda3\lib\site-
packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in c:\anaconda3\lib\
site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\anaconda3\lib\
site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\anaconda3\lib\site-
packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\anaconda3\lib\
site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\anaconda3\lib\
site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in c:\anaconda3\lib\
site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in c:\anaconda3\lib\site-
packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\anaconda3\lib\
site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in c:\anaconda3\lib\site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
df = pd.read_csv('customer churn.csv')
df
```

```
      customerID  gender  SeniorCitizen Partner Dependents  tenure  \
0     7590-VHVEG  Female              0     Yes         No       1
1     5575-GNVDE    Male              0      No         No      34
2     3668-QPYBK    Male              0      No         No       2
3     7795-CFOCW    Male              0      No         No      45
4     9237-HQITU  Female              0      No         No       2
...          ...     ...            ...     ...        ...     ...
7038  6840-RESVB    Male              0     Yes        Yes      24
7039  2234-XADUH  Female              0     Yes        Yes      72
```

```
7040  4801-JZAZL  Female              0     Yes        Yes      11
7041  8361-LTMKD    Male              1     Yes         No       4
7042  3186-AJIEK    Male              0      No         No      66

     PhoneService     MultipleLines InternetService
OnlineSecurity  ...  \
0              No  No phone service             DSL
No  ...
1             Yes                No             DSL
Yes  ...
2             Yes                No             DSL
Yes  ...
3              No  No phone service             DSL
Yes  ...
4             Yes                No     Fiber optic
No  ...
...           ...               ...             ...             ...  ..
.
7038          Yes               Yes             DSL
Yes  ...
7039          Yes               Yes     Fiber optic
No  ...
7040           No  No phone service             DSL
Yes  ...
7041          Yes               Yes     Fiber optic
No  ...
7042          Yes                No     Fiber optic
Yes  ...

     DeviceProtection TechSupport StreamingTV StreamingMovies
Contract  \
0                  No          No          No              No  Month-
to-month
1                 Yes          No          No              No
One year
2                  No          No          No              No  Month-
to-month
3                 Yes         Yes          No              No
One year
4                  No          No          No              No  Month-
to-month
...               ...         ...         ...             ...
...
7038              Yes         Yes         Yes             Yes
One year
7039              Yes          No         Yes             Yes
One year
7040               No          No          No              No  Month-
to-month
```

```
7041                No          No          No               No  Month-
to-month
7042               Yes         Yes         Yes              Yes
Two year

     PaperlessBilling              PaymentMethod MonthlyCharges
TotalCharges  \
0                 Yes           Electronic check          29.85
29.85
1                  No               Mailed check          56.95
1889.5
2                 Yes               Mailed check          53.85
108.15
3                  No  Bank transfer (automatic)          42.30
1840.75
4                 Yes           Electronic check          70.70
151.65
...               ...                        ...            ...
...
7038              Yes               Mailed check          84.80
1990.5
7039              Yes    Credit card (automatic)         103.20
7362.9
7040              Yes           Electronic check          29.60
346.45
7041              Yes               Mailed check          74.40
306.6
7042              Yes  Bank transfer (automatic)         105.65
6844.5

     Churn
0       No
1       No
2      Yes
3       No
4      Yes
...    ...
7038    No
7039    No
7040    No
7041   Yes
7042    No

[7043 rows x 21 columns]
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
```

```
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

#replacing blanks with 0 as tenure is 0 and no total charges are recorded

```
df["TotalCharges"] = df["TotalCharges"].replace(" ","0")
df["TotalCharges"] = df["TotalCharges"].astype("float")

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
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 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
```

```
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   float64
 20  Churn             7043 non-null   object
dtypes: float64(2), int64(2), object(17)
memory usage: 1.1+ MB
```

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges
count	7043.000000	7043.000000	7043.000000	7043.000000
mean	0.162147	32.371149	64.761692	2279.734304
std	0.368612	24.559481	30.090047	2266.794470
min	0.000000	0.000000	18.250000	0.000000
25%	0.000000	9.000000	35.500000	398.550000
50%	0.000000	29.000000	70.350000	1394.550000
75%	0.000000	55.000000	89.850000	3786.600000
max	1.000000	72.000000	118.750000	8684.800000

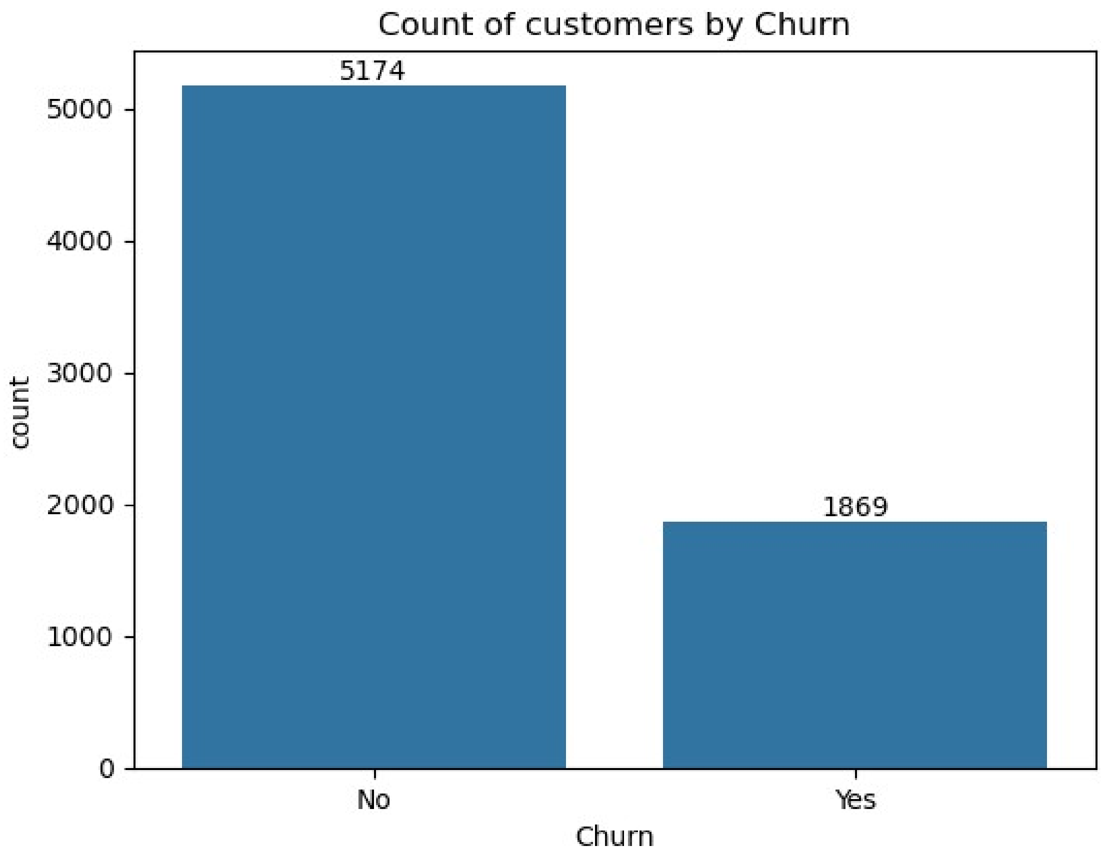
```
df["customerID"].duplicated().sum()
```

```
np.int64(0)
```

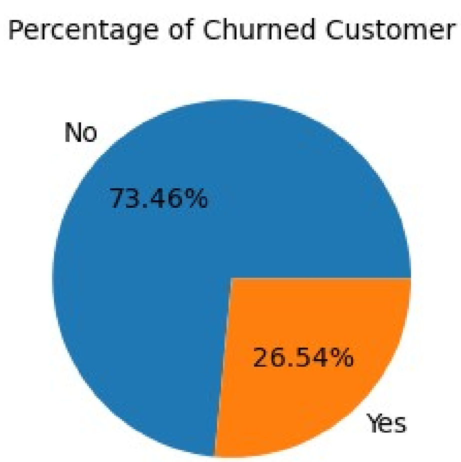
```
def conv(value):
    if value == 1:
        return "yes"
    else:
        return "no"

df['SeniorCitizen'] = df["SeniorCitizen"].apply(conv)
```

#converted 0 and 1 values of senior citizen to yes/no

```
ax = sns.countplot(x = 'Churn', data = df)
ax.bar_label(ax.containers[0])
plt.title("Count of customers by Churn")
plt.show()
```

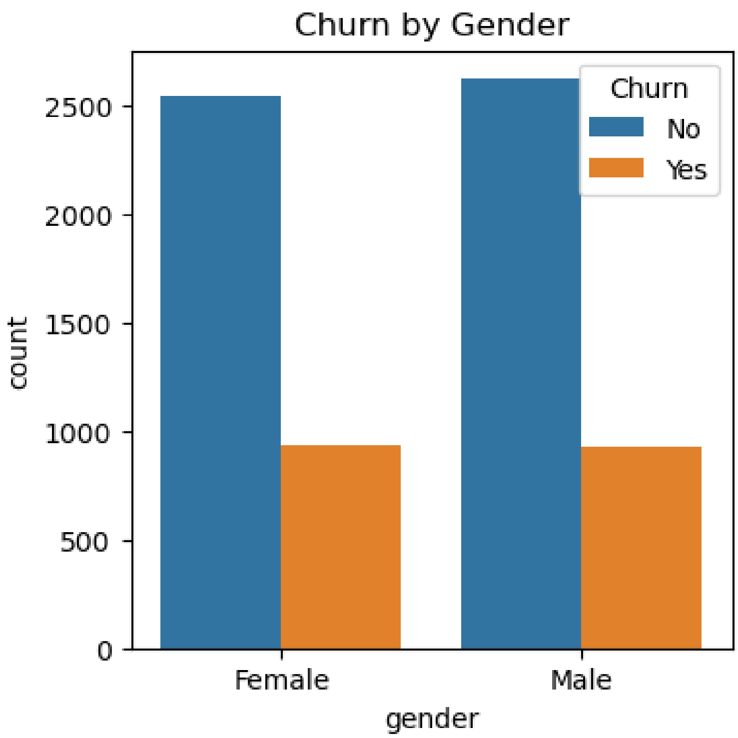



```python
plt.figure(figsize = (3,3))
gb = df.groupby("Churn").agg({'Churn' : "count"})
plt.pie(gb['Churn'], labels = gb.index, autopct = "%1.2f%%")
plt.title("Percentage of Churned Customer", fontsize = 10)
plt.show()
```

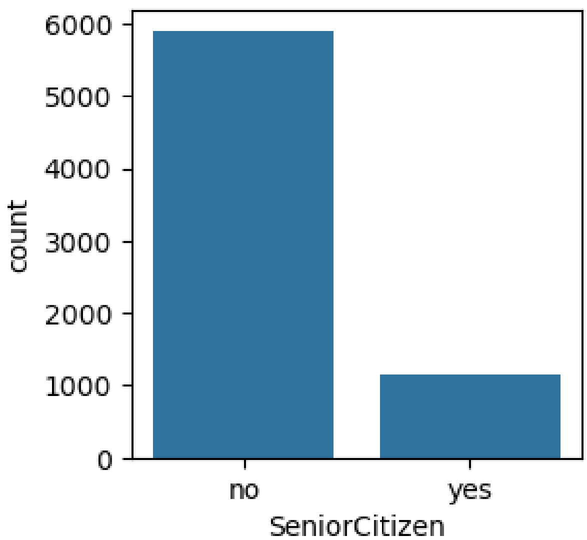


#from the given pie chart we can conclude that 26.54% of ourcustomers have churned out.#now let's explore the reason behind it

```
plt.figure(figsize = (4,4))
sns.countplot(x = "gender", data = df, hue = "Churn")
plt.title("Churn by Gender")
plt.show()
```




```
plt.figure(figsize = (3,3))
ax = sns.countplot(x = "SeniorCitizen", data = df)
ax.bar_label(ax.container[0])
plt.title("Count of Customers by SeniorCitizen")
plt.show()

----------------------------------------------------------------------
-----
AttributeError                            Traceback (most recent call
last)
Cell In[14], line 3
      1 plt.figure(figsize = (3,3))
      2 ax = sns.countplot(x = "SeniorCitizen", data = df)
----> 3 ax.bar_label(ax.container[0])
      4 plt.title("Count of Customers by SeniorCitizen")
      5 plt.show()

AttributeError: 'Axes' object has no attribute 'container'
```

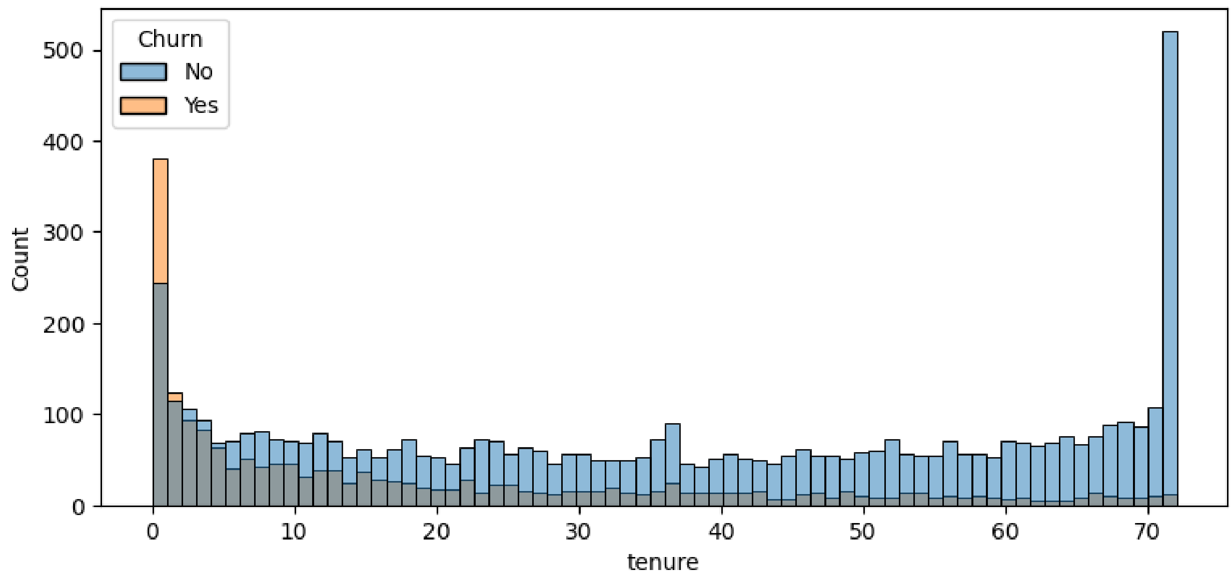



```
total_counts = df.groupby('SeniorCitizen')
['Churn'].value_counts(normalize=True).unstack()

# Create figure
fig, ax = plt.subplots(figsize=(6,6))

# Plot stacked bar chart
total_counts.plot(
    kind='bar',
    stacked=True,
    ax=ax,
    color=['#1f77b4', '#ff7f0e']
)

# Add percentage labels
for p in ax.patches:
    width = p.get_width()
    height = p.get_height()
    x, y = p.get_xy()

    ax.text(
        x + width / 2,
        y + height / 2,
        f'{height:.1%}',
        ha='center',
        va='center'
    )

# Titles and labels
plt.title('Churn by Senior Citizen (Stacked Bar Chart)')
plt.xlabel('SeniorCitizen')
plt.ylabel('Percentage')
```

```
plt.xticks(rotation=0)

# Legend
plt.legend(title='Churn', bbox_to_anchor = (0.9,0.9))

plt.show()
```

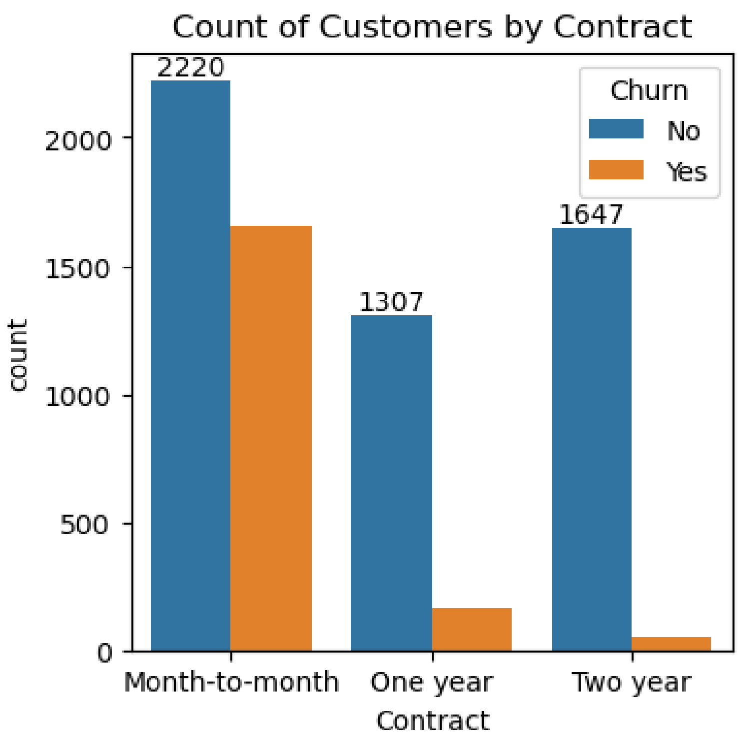
#compartive a greater precentage of people in senior citizen category have churned

```
plt.figure(figsize = (9,4))
sns.histplot(x = "tenure", data = df, bins = 70, hue = "Churn")
plt.show()
```



people who used our services for a long time have stayed and people who have used for 2 months

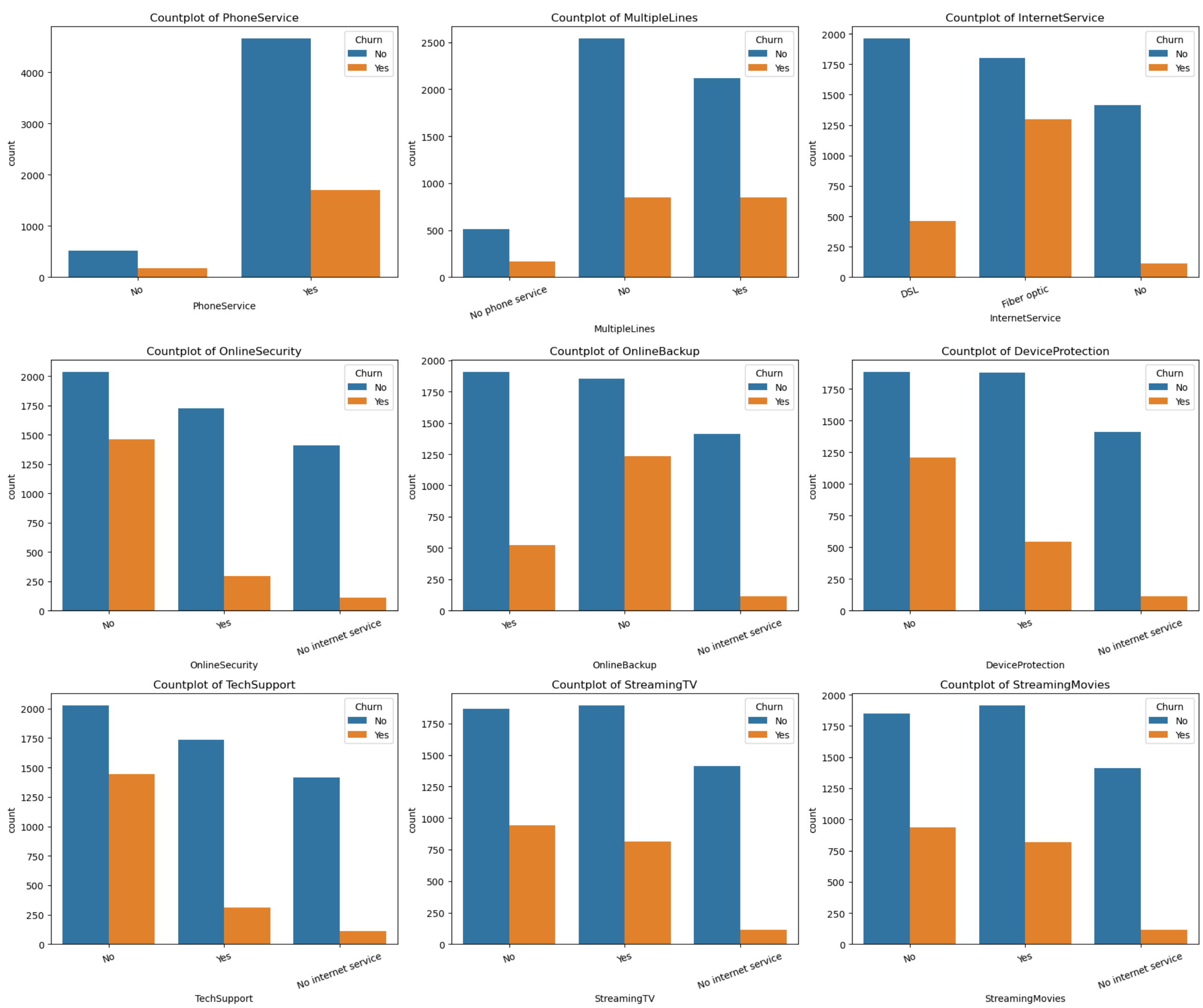
```
plt.figure(figsize = (4,4))
ax = sns.countplot(x = "Contract", data =df, hue = "Churn")
ax.bar_label(ax.containers[0])
plt.title("Count of Customers by Contract")
plt.show()
```



#poeple who have month contract are likely to churn then from those who have 1 or 2 years of contract

```
df.columns.values

array(['customerID', 'gender', 'SeniorCitizen', 'Partner',
'Dependents',
       'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
       'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
       'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract',
       'PaperlessBilling', 'PaymentMethod', 'MonthlyCharges',
       'TotalCharges', 'Churn'], dtype=object)

columns = [
    'PhoneService', 'MultipleLines', 'InternetService',
    'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
    'TechSupport', 'StreamingTV', 'StreamingMovies'
]

# Create subplot layout
fig, axes = plt.subplots(nrows=3, ncols=3, figsize=(18, 15))

# Flatten axes for easy looping
axes = axes.flatten()

# Loop through columns
for i, col in enumerate(columns):
```

```
    sns.countplot(x=col, data=df, ax=axes[i], hue = df["Churn"])

    # Title for each subplot
    axes[i].set_title(f'Countplot of {col}')

    # Rotate labels if needed
    axes[i].tick_params(axis='x', rotation=20)

# Adjust layout
plt.tight_layout()

# Show plots
plt.show()
```



#The majority of customers who do not churn tend to have services like PhoneService, InternetService (particularly DSL), and OnlineSecurity enabled. For services like OnlineBackup, TechSupport, and StreamingTV, churn rates are noticeably higher when these services are not used or are unavailable

```
plt.figure(figsize = (6,4))
ax = sns.countplot(x = "PaymentMethod", data = df, hue = "Churn")
ax.bar_label(ax.containers[0])
ax.bar_label(ax.containers[1])
plt.title("Churned Customers by Payment Method")
plt.xticks(rotation = 45)
plt.show()
```